

Lab Worksheet

PC Component Analysis

Computer Use

E.g. gaming, multimedia creation, general, etc.

Machine Learning Research PC, can be better by looking for retired server parts such as MI150 GPU or its siblings for lots of HBM2 memory for sub \$1000 cost.

Choose your Components

Component	Function	Key Specifications	Justification
CPU	Processes instructions	4.5 GHz / 5.7 GHz	High core count
GPU	Processes high bandwidth instructions	32GB HBM2	High memory and speed
RAM	Hold temporary values for system	96GB	Needed for managing bifurcated models when exceeding video memory, cpu or ram should be swapped for frequency matching
Motherboard	The substrate that connects all components	PCIE GEN 5, AM5	Full size to reduce heat concentration and provide ability for future expansion
Storage	Non volatile memory for holding permanent files	1TB ssd 10TB HDD	Fast storage for system and immediate data, large storage for training sets
PSU	Power supply	1000w Modular	High capacity to optimize the efficiency curve given hardware power draw

Case	Holds all the good stuff	Full ATX	Good air flow and space for large GPUs
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Link to build

<https://buildmypc.net/rigs/eUFU3im/>